1 Solution Architecture

The solution enforces data integrity in the browser, at the point of entry. A four-layer architecture keeps client-side controls (UI Policies and Client Scripts) at the centre, backed by platform services and the data layer, and delivered through standard ServiceNow development tooling.

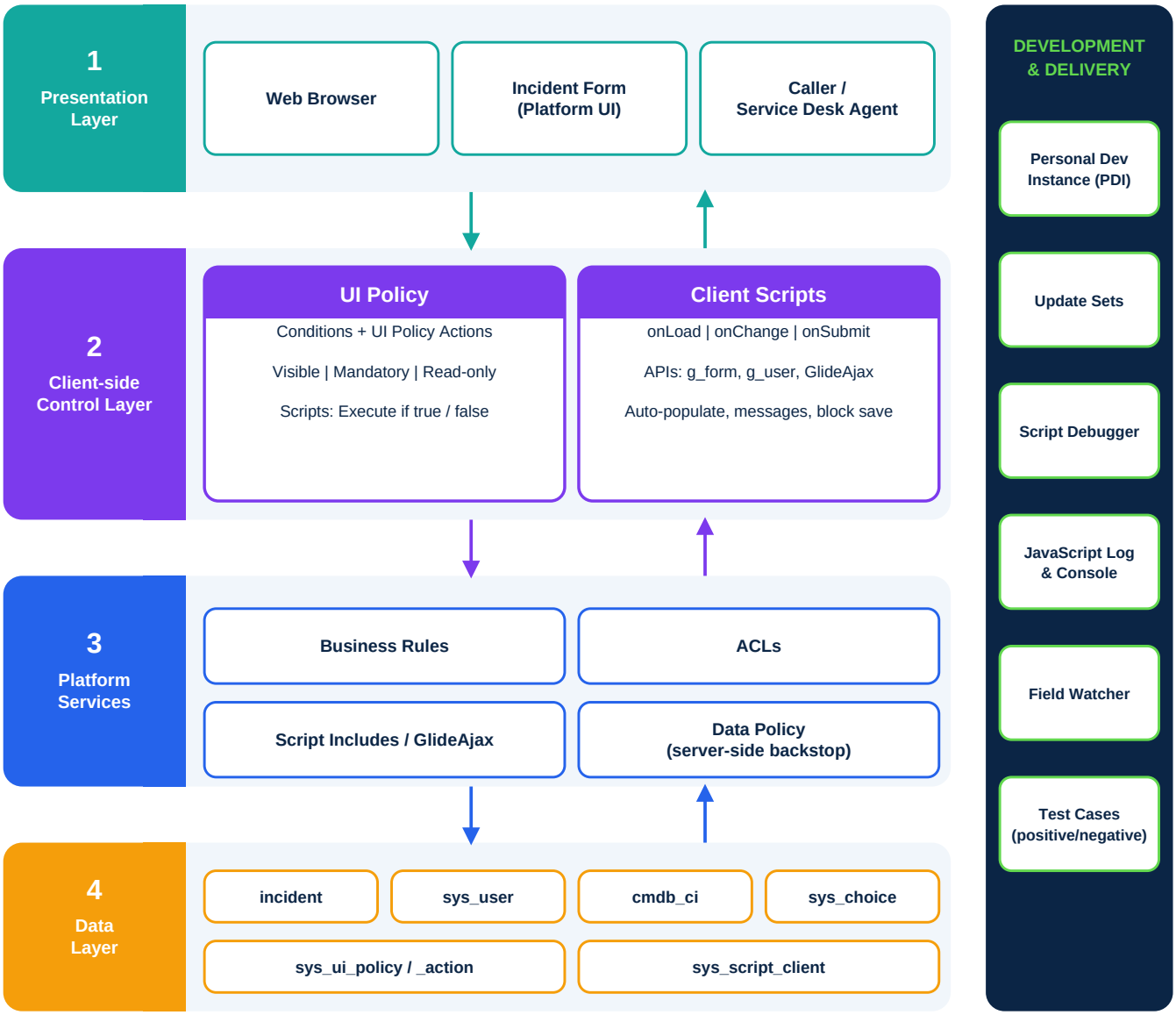

Figure 1: Layered architecture of the Incident client-side validation solution

2 Technology Stack

Layer	Technology	Purpose
Platform	ServiceNow (Now Platform)	Hosts the Incident application, forms, rules and the database.
Application	Incident Management (incident table)	Process and table that receive the data being validated.
Client-side control (no-code)	UI Policy and UI Policy Actions	Condition-based visible, mandatory and read-only behaviour on the form.
Client-side control (script)	Client Scripts: onLoad, onChange, onSubmit	Auto-populate values, show messages, clear dependent fields, block save.
Language	JavaScript (client-side)	Logic inside UI Policy scripts and Client Scripts.
Client APIs	g_form, g_user, GlideAjax	Read and set fields, set mandatory / visible / read-only, show messages, call server code.
Server support	Script Includes, Business Rules, ACLs, Data Policy	Server-side lookups and a safety net for rules that must never be bypassed.
Data	incident, sys_user, cmdb_ci, sys_choice	Record data and reference data for auto-population.
Configuration records	sys_ui_policy, sys_ui_policy_action, sys_script_client	Where the rules themselves are stored.
Delivery	Update Sets, Personal Developer Instance	Build, capture and move the configuration between instances.
Testing and debugging	Script Debugger, JavaScript Log, Field Watcher, browser DevTools	Verify rule behaviour and trace field changes.